Scalability & Future Plan

Date	28 October 2026
Team ID	[Insert Your Actual Team ID, e.g., TM-1042]
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Cross-sectional dataset size restricted to static 5,000 synthetic patient records without real-time EHR stream ingestion.	Prevents real-time live monitoring of acute patient vital fluctuations in active emergency departments.	High
2	Rule-based static risk tiering logic ([Risk Tier Index]) instead of predictive Machine Learning classifiers.	Limits automated early warning detection for 30-day hospital readmissions or acute heart disease onset.	Medium
3	Local Flask web server hosting (http://127.0.0.1:5001) without multi-tenant authentication or role-based access control (RBAC).	Restricts secure enterprise deployment across separate hospital departments or regional clinic networks.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local presentation mode on desktop environment.	Deploy Flask application onto AWS Elastic Beanstalk with Auto-Scaling Groups and NGINX reverse proxy load balancing.
2	Data Storage	Static CSV batch staging in Snowflake X-Small virtual warehouse (HEALTH360_WH).	Migrate to Snowflake Dynamic Tables and automated Snowpipe streaming for auto-scaling data ingestion pipelines.
3	Performance	~345ms Snowflake query execution and sub-second Tableau client-side filtering.	Implement Tableau Server / Online Extract API pre-caching and Snowflake materialized clustering keys.
4	Security	Unauthenticated local web shell container.	Implement OAuth2 / SAML 2.0 single sign-on (SSO), HIPAA-compliant SSL/TLS encryption, and column-level data masking.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Machine Learning Readmission Prediction (XGBoost / Random Forest) integrated into Flask API backend.	Months 1–3	Provides clinical teams with 88%+ predictive accuracy for 30-day readmission risks.
Phase 2	HL7 / FHIR API integration with hospital Electronic Health Record (EHR) systems (Epic / Cerner).	Months 4–6	Enables real-time automated ingestion of live patient vitals and ICU bed availability metrics.
Phase 3	IoT Wearable Vitals Streaming Engine (Continuous Glucose Monitors & Smartwatches) and Mobile Health App (iOS/Android).	Months 7–12	Extends Health360 beyond hospital walls for proactive remote outpatient monitoring and community health management.